

BIKANER, India

WMO: 421650

Lat: 28.0207N

Lon: 73.2772E

Elev: 224

StdP: 98.66

Time Zone: 5.50 (IND)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.2	(c) 7.6	(d) -8.4	(e) 1.9	(f) 22.2	(g) -5.8	(h) 2.4	(i) 22.0	(j) 4.6	(k) 18.4	(l) 3.4	(m) 17.3	(n) 0.4	(o) 50	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 10.6	(c) 44.1	(d) 21.6	(e) 42.8	(f) 22.0	(g) 41.6	(h) 22.4	(i) 28.1	(j) 34.5	(k) 27.6	(l) 33.9	(m) 27.1	(n) 33.5	(o) 2.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.9	30.5	26.2	22.2	30.3	25.7	21.5	30.3	92.5	34.4	89.9	33.8	87.8	33.5	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.6	(b) 5.2	(c) 4.1	(e) 2.9	(f) 45.9	(g) 1.7	(h) 1.3	(i) 1.7	(j) 46.8	(k) 0.8	(l) 47.6	(m) -0.2	(n) 48.3	(o) -1.4	(p) 49.3		
(a) 6.6	(b) 5.2	(c) 4.1	(e) 0.9	(f) 29.2	(g) 1.3	(h) 0.9	(i) 0.0	(j) 29.9	(k) -0.8	(l) 30.4	(m) -1.5	(n) 30.9	(o) -2.5	(p) 31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	15.5	19.7	25.6	31.1	35.1	35.5	33.5	32.0	31.3	28.6	23.0	17.3
	DBStd	7.23	2.87	3.24	3.66	3.03	2.65	2.37	2.20	1.72	1.77	2.55	2.79	2.97
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	176	97	20	1	0	0	0	0	0	0	0	3	55
	CDD10.0	6349	171	272	485	632	778	764	729	683	639	577	391	228
	CDD18.3	3482	8	59	228	382	520	514	471	425	389	319	143	24
	CDH23.3	51161	85	566	2647	5698	8794	8696	7363	6271	5529	3935	1344	232
	CDH26.7	32634	7	177	1389	3651	6365	6317	4901	3805	3229	2199	553	41
(14) Wind	WSAvg	1.2	0.7	0.9	1.1	1.3	1.7	2.1	1.9	1.4	1.1	0.7	0.5	0.5
(15) Precipitation	PrecAvg	296	4	9	8	6	24	40	96	62	34	10	2	2
	PrecMax	622	28	104	62	52	95	202	372	202	198	128	42	42
	PrecMin	55	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	123	6	18	14	12	27	40	80	44	41	24	6	6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.7	33.7	39.2	43.2	46.0	45.6	42.7	40.3	40.4	39.1	34.2	29.7
		MCWB	14.6	16.7	17.9	19.8	20.9	21.9	25.7	26.1	22.9	19.9	18.0	16.4
	2%	DB	25.7	30.8	37.3	41.6	44.7	43.7	41.0	38.6	39.0	37.7	32.7	27.9
		MCWB	13.7	16.3	17.7	19.4	21.0	22.8	25.5	26.0	22.9	20.8	17.8	15.7
	5%	DB	24.0	28.9	35.5	40.2	43.5	42.5	39.7	37.5	37.7	36.4	31.5	26.2
		MCWB	13.2	15.9	17.4	19.2	21.0	23.5	25.7	25.7	23.5	20.5	17.6	14.7
	10%	DB	22.4	26.9	33.5	38.7	42.1	41.2	38.5	36.5	36.4	35.4	30.1	24.3
		MCWB	12.4	14.9	16.9	19.0	21.3	23.8	25.7	25.7	23.8	20.2	17.1	13.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	20.7	21.4	24.0	26.8	28.5	28.8	28.8	28.5	25.5	21.7	18.5
		MCDB	23.2	27.2	31.6	37.9	38.1	37.7	35.5	34.2	34.4	33.1	29.9	26.5
	2%	WB	15.4	18.7	19.9	22.0	25.5	27.2	28.0	27.5	27.5	24.3	20.0	17.1
		MCDB	22.2	26.1	29.9	35.4	36.5	36.0	34.2	33.9	33.5	32.0	29.3	24.8
	5%	WB	14.3	17.4	19.0	21.0	24.7	26.5	27.5	27.4	26.8	23.3	18.6	15.9
		MCDB	21.2	25.3	30.0	35.0	36.0	36.0	33.9	33.2	32.6	31.4	29.0	23.1
	10%	WB	13.3	16.3	18.2	20.1	24.0	26.0	27.1	27.0	26.2	22.1	17.7	14.7
		MCDB	20.6	23.9	29.2	34.6	35.7	35.9	33.6	32.7	31.6	31.6	28.2	22.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	12.8	13.2	13.4	12.7	10.6	8.9	8.4	9.8	12.8	14.1	13.7
		MCDBR	14.4	14.6	14.7	14.4	13.8	11.7	10.1	9.6	11.4	13.5	14.7	15.1
	5% WB	MCWBR	6.8	5.9	5.0	5.0	4.6	3.6	2.3	2.4	3.4	4.6	6.3	6.9
		MCWBR	12.4	12.2	12.6	12.7	12.0	10.0	9.1	8.3	8.8	11.6	13.5	13.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.432	0.438	0.479	0.561	0.643	0.716	0.740	0.641	0.548	0.540	0.502	0.457
		taud	1.970	1.997	1.900	1.741	1.614	1.539	1.531	1.719	1.876	1.780	1.816	1.906
	Ebn at Noon	Ebn at Noon	782	821	812	759	698	646	628	690	743	711	702	730
		Edh at Noon	156	164	192	232	264	284	285	235	195	202	181	160
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.84	4.84	6.03	6.82	7.09	6.58	5.98	5.83	5.82	5.26	4.15	3.67
	RadStd	RadStd	0.20	0.20	0.22	0.19	0.15	0.30	0.30	0.36	0.28	0.20	0.22	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page